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COS700 Research Proposal

A Digital Forensics Readiness Model For AI-Powered Cyber Attacks

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Abstract

Digital forensics is the study of scientifically proven methods that result in the identification, preservation, collection, validation, analysis, interpretation, documentation, and presentation of digital evidence. As Artificial Intelligence (AI) is increasingly used in our daily lives, it is also being incorporated into cybersecurity. Although using AI in cybersecurity has its positive results, it is also being weaponized for cyber-attacks. The current or rather traditional digital forensics readiness models cannot keep up with the evolving AI-driven cyber-attacks due to how complex they are becoming. This research proposes a Digital Forensics Readiness Model specifically made for AI-powered attacks, the research aims to address the current limitations of the models it also seeks to bridge the gap between developing cyber-attacks/threats and the preparedness of digital forensics.

Keywords: Digital Forensics, Cybersecurity, Artificial Intelligence, Threat detection, Anomaly detection, Forensic detection

Contents

1	Introduction	4
2	Problem Statement	5
2.1	Research Question:	5
2.1.1	What are the current Digital Forensics Readiness Models that are being used?	5
2.1.2	What role can block-chains play into fighting and detecting cyber-attacks	6
2.1.3	How can machine learning and AI forensics techniques assist with improving real time detection	6
2.2	Motivation	6
2.3	Aims and Limitations	6
3	Literature Study	8
3.1	What is digital forensics?	8
3.2	The role of machine learning in digital forensics	8
3.3	Existing Digital Forensics Readiness Models	8
3.3.1	Kruse Heiser Model (Lucent Model)	8
3.3.2	United States Department of Justice (USDOJ) model	9
3.3.3	The Digital Forensic Research Workshops (DFRWS) model	9
3.3.4	Abstract Digital Forensics Model (ADFM)	9
3.3.5	Integrated Digital Investigation Process (IDIP)	10
3.3.6	Systemic Digital Forensic Investigation(SDFIM) Model	10
3.4	The role of block-chains in digital forensics	10
3.4.1	AI-Powered cyber-attacks and how they impact the digital forensics space.	11
4	Methodology	12
5	Planning	13
6	References	14

1 Introduction

Experts say that artificial intelligence will fuel cybercrime attacks in the current year 2025 [3]. As defined by the Oxford Dictionary, artificial intelligence (AI) involves the use of computer systems to carry out tasks or generate outputs that typically require human intelligence, often through the application of machine learning methods to extensive datasets. While AI has its own advantages and good uses, as it rapidly grows, there are criminals that use it for what we call cybercrime [4].

Artificial intelligence is being used by criminals for automated, adaptive and large-scale threats [5] which can also be known as AI-Powered cyber attacks. While AI-powered security solutions help companies detect and prevent cyber-attacks, hackers are also using them to develop advanced attack techniques such as deepfake-based fraud [5]. Which is when an AI tool is used to generate fake media that manipulates facial features. An example is when they use another person's video where they are saying a completely different thing and change it to what they want that person to say, usually asking for money.

They also use it for phishing schemes which is the most common. This is when they trick you into releasing very sensitive information. An example would be an SMS (short message service) stating that you received money, but to claim you need click on the link. Then before you know it your money is gone. It is reported that globally 57 percent of companies receive phishing attempts [6].

Lastly, these criminals use it for autonomous malware[5]. Which is a type of attack that can execute without any human intervention (no clicking links or anything of that nature). It follows a predefined set of instructions. It is actually one of the hardest to control or even detect because it has the ability to act independently.

Therefore, the use of AI both as a defensive tool and a dangerous weapon poses significant challenges and threats for digital forensics. To counteract these attacks, digital forensic readiness models are being used. The goal of the models is to detect the cyber-attacks before they occur or create a pattern for better detection and prevention.

One of the ways they do this is by creating a blockchain, which is a distributed, decentralized ledger technology that securely records transactions across computers. These blockchains ensure integrity in the evidence, as they cannot be altered with, making it trustworthy and the evidence to have integrity. Blockchains enhance digital forensic investigations.

Despite the existence of the current digital forensics readiness models named: Systemic Digital Forensic Investigation (SDFIM) Model, Kruse Heiser Model also known as Lucent, United States Department of Justice (USDOJ) model, Abstract Digital Forensics Model (ADFM), The Integrated Digital Investigation Process (IDIP), Digital Forensic Research Workshop model, and United States Department of Justice (USDOJ) model [7], [8], [9], many of them are currently not working as effectively as they should.

This is mainly due to the fast-evolving threat of artificial intelligence making the field of digital forensics quite challenging especially due to AI-Powered cyber-attacks/threats.

The purpose of this research is to create a digital readiness forensics model that can counteract the cyber-attacks, this will be done by looking at the current or rather traditional methods that are being used and address the factors that make them not so effective as they should be. One of the features the model will entail is the ability to have real time detection/response time.

The rest of the research proposal is structured as described below. The next section will go into the problem statement, which will be followed by the literature study and then after the research methodology and lastly the project plan.

2 Problem Statement

As artificial intelligence keeps evolving, it has some positive impacts especially in the field's cybersecurity and digital forensics. While the growth of AI has positive impacts it also has some negative aspects mainly due to the fact that certain developers use it for malicious purposes. The main concerns being deepfake fraud/phishing as well as autonomous malware. The increasing use of AI-Powered cyber-attacks using deepfake based phishing, or autonomous malware makes it harder for traditional or current digital forensic readiness models to 'keep up'. The current models are struggling to effectively detect, analyze and investigate the cyber-attacks. Which is why it is quite important for these models to evolve for them to be able to counterattack them. The objective of the research is to develop an evolved Digital Forensic Readiness Model that comprises of AI-Powered forensic techniques which have the ability to detect cyber-attacks in real time.

2.1 Research Question:

The objective of this research is to investigate how digital forensics readiness models can be improved or evolved to effectively detect, analyze as well as respond to AI-Powered cyber-attacks in real time.

2.1.1 What are the current Digital Forensics Readiness Models that are being used?

This question aims to analyze and address the current or existing digital forensics readiness models. The aim is to analyze the type of techniques they use. This research will also investigate what is currently the problem of the models not being able to effectively detect and respond to the cyber-attacks. This research will also dive into how these models can be evolved or improved for them

to work effectively.

2.1.2 What role can block-chains play into fighting and detecting cyber-attacks

This question aims to analyze and research current block-chain technologies that are being used and how they can be used in digital forensics and cybersecurity. This research will go through the current block-chains that are being used and how they are helpful during cyber-attacks. This research will also explore how block-chains ensure and maintain data integrity as well as traceability. Lastly it will also research the limitations and challenges of using block-chains in digital forensics.

2.1.3 How can machine learning and AI forensics techniques assist with improving real time detection

This question aims to research about the current AI-driven forensic techniques that are currently being used. It will also research on how machine learning algorithms can identify AI-Powered cyber threats or attacks more efficiently. This research will also dive into the pros of machine learning and AI in digital forensics. Lastly, the research will dive more into the risks and challenges that are associated with using machine learning and AI forensics techniques.

2.2 Motivation

The main motivation of the research proposal is the rather increment of AI-Powered cyber-attacks and the less effective digital forensics readiness models. Especially when it comes to these evolving and growing threats. The existing models struggle to keep up with the advancement of AI, they lack real-time detection which is essential. This research aims to bridge the gap by introducing a Digital Forensics Model specifically made for AI-Powered cyber-attacks. Which will entail real-time detection and make use of blockchain based evidence logging. This will make the model more effective and more advanced than the current existing models.

2.3 Aims and Limitations

The main aim of this research is to create a Digital Forensics Readiness Model for AI-Powered cyber-attacks. The model will entail the enhancement of detection, analysis and the way it responds to attacks such as deepfake fraud, AI-Powered phishing and autonomous malware. The aim is to incorporate real-time detection and use blockchains as a way of evidence logging. This model aims to decrease the AI-Powered cyber-attacks.

The limitation of this research is the fact that the model may not be able to eliminate all the AI-Powered cyber-attacks as these AI advancements tend to happen all the time. It is almost impossible to generate a model that will be able to detect even future attacks. This is mainly due to the fact that we simply do not know what AI has in store for the future.

3 Literature Study

The following sections are intended to present and provide a brief overview of the research paper's content. The first section explains what digital forensics is as well as how machine learning plays a role in digital forensics, then followed by the descriptions of the existing digital forensics models that are being used along with the gaps that are identified. The following section will be explaining the role of block-chains in digital forensics, and explain AI-Powered cyber-attacks and how they impact the digital forensics space.

3.1 What is digital forensics?

Digital forensics is the discipline focused on preserving, analyzing, presenting, and identifying information and evidence from electronic and digital devices while safeguarding the privacy of users [10]. One of the main objectives, or rather purposes of digital forensics is to investigate cybercrime. They gather evidence and patterns that can be used in court. As mentioned prior as AI develops, so are cyber crimes. It is as important to be able to catch these criminals, and more importantly send them to jail. There are different types of digital forensics readiness models of which we will dive into in the next subsections. The purpose of these models is to detect, respond to, and investigate.

3.2 The role of machine learning in digital forensics

By including machine learning in digital forensics it makes the whole investigation process much easier because machine learning algorithms are able to learn the abnormalities of a system. If there is a breach in the system it instantly detects it. It is also able to detect deepfake phishing as well as identifying malware.

3.3 Existing Digital Forensics Readiness Models

3.3.1 Kruse Heiser Model (Lucent Model)

Kruse and Heiser Model is one of the earliest known methodological approaches that was introduced to digital forensics. The most important aspect of this model is keeping its data integrity as well as its validity. It is separated into three sections, 1. Getting evidence without changing them in any way. 2. Verifying evidence to ensure that the data is consistent with the data originally gathered/taken. Lastly, analyzing the evidence/data without tampering it in any way. The model being one of the oldest and one of the simplest models implies it has some limitations. [7]. The model itself has the disadvantage of having struggles or rather limitations of presenting the evidence to the law.

The model is also less proactive as it only investigates the cyber attack once it has already happened. The type of cyber-attacks that we are experiencing now require real-time detection and need a proactive model rather than a reactive one. The model was created for the purposes of detecting crimes such as unauthorized access or data breaches. Therefore, when it comes to AI-Powered cyber-attacks such as deep-fake fraud, this model does not work because it lacks the capabilities.

3.3.2 United States Department of Justice (USDOJ) model

The United States Department of Justice is a 4 step phased model, which comprises of examination, collection, analysis and lastly reporting. The Collection phase deals with the collection and identification of digital evidence[7]. It also ensures that the integrity of the data is kept and value of the evidence is not changed or tampered with. The phase that follows which is the examination phase deals with performing the retrieving or fetching the digital evidence from the collected evidence. The second last phase is analysis which deals with analyzing the collected data or evidence. This is the phase where it deals with identifying certain patterns, connections and anomalies that can lead to the cyber attacker or rather indicate of a cyber-attack. The last phase in this model is the reporting phase. This is where the evidence found is documented so that they can be used during court cases or during criminal investigations.

3.3.3 The Digital Forensic Research Workshops (DFRWS) model

This model provides a consistent standard forensic framework. The model starts with the identification process, this is where profiles are detected, as well as systems are observed and an audit evaluation is carried out. [7]. Which is then followed by the preservation, which includes having a structured case management and making ensuring the chain of evidence is acceptable. The preservation phase is one of the most important or rather crucial phases because it is where the model makes sure that the data has not been tampered with or changed which ensures its integrity is maintained. The phase that follows is collection which is where important data are collected or gathered using various recovery techniques. Following the above phases, examination and analysis phases follow, which is where evidence tracing, evidence validation as well as the recovery of encrypted data is performed. Lastly, presentation phase is where all the data and evidence are documented and presented.

3.3.4 Abstract Digital Forensics Model (ADFM)

This model is an extension of the Digital Forensics Research Workshops (DFRWS). The model includes all the phases of DFRWS but has an extension of 3 more additional phases/sections, making the total number of phases/sections to 9. The phases added were, the approach strategy, preparation and returning evidence[7]. The preparation phase involves dealing with tools and technique

preparation, which includes search warrants and monitoring authorizations to further the investigation. The approach strategy phase involves starting with the aim to maximize the collection of unaltered data while trying to lessen the impact to the victim. The last additional phase being returning evidence which makes sure that the physical and digital property is returned to the rightful owner and determines which criminal evidence needs to be removed. The main advantage of this model is the fact that it is quite diverse, it can be used by observers with no experience in the field as it is user friendly. One drawback is the lack of a straightforward method for testing the model.

3.3.5 Integrated Digital Investigation Process (IDIP)

IDIP was organized into 5 groups which contain 17 phases. The groups are deployment phase, readiness phase, digital crime scene investigation phase, physical crime scene investigation phase and lastly review phase[7]. The readiness phase focuses on the operational and infrastructural support. The deployment phase detects and confirms an incident, it can be split into detection and authorization phase. The gathering of physical evidence is done at an actual crime scene investigation. The digital crime scene investigation follows the phases preservation, survey, documentation, searching and collection of evidence. The last phase in IDIP is the review phase in which the evidence is reconstructed and therefore presented in court.

3.3.6 Systemic Digital Forensic Investigation(SDFIM) Model

SDFIM has 11 phases named securing the scene, survey and recognition, communication shielding, preparation, examination, documentation of scene, evidence collection, preservation, analysis, presentation, review and result. This model is the most flexible and more practical, it is divided into 11 phases in which each step loops back to add more flexibility[7]. The main con of this model is that since there are so many phases, this makes it quite complex and more time consuming.

3.4 The role of block-chains in digital forensics

In digital forensics it is very critical to ensure that evidence is not tampered with. Any alteration to the evidence can make the whole case dismissed. This is then where block-chains come in, they ensure that the evidence remains trustworthy and can be verified during investigations. With block-chains evidence or data cannot be altered, it keeps a train of evidence. Every single entry or addition to evidence is recorded and it shows who it was done by. In simple terms it also acts as a log book. As cyber threats become more complicated and complex, the use of block-chains in digital forensics strengthens the credibility and security of the collection of evidence. Investigators are able to use it as evidence in front of the court of law [12].

3.4.1 AI-Powered cyber-attacks and how they impact the digital forensics space.

Unlike traditional attacks, AI-Powered are intelligent, adaptive, faster and have the capability of learning from their environment. Which automatically makes them so much harder to detect and investigate. The traditional methods are not equipped enough to handle these attacks and the rate of which they are growing. Due to the attacks, evidence is even harder to trace, data is manipulated even more easily. Furthermore, the use of artificial intelligence by these attackers challenges the foundation of forensic analysis which strongly rely on patterns and anomalies. It makes it much harder for forensic investigators to be able to detect/identify malicious activities [11].

The next section will further explain the methodologies that will be used during the research project.

4 Methodology

The research methodology used will be a combination of qualitative methodology as well as quantitative method, which will help with effectively analyzing digital forensics readiness models for AI-Powered cyber-attacks. The first section of the methodology will be a literature review, where the analyzing of existing digital forensic readiness models will occur. After that, the research will investigate blockchain techniques for AI-Powered forensic techniques as well as identify the current limitations and current gaps that the readiness models have. To do this, the research will review sources such as journals, books, academic papers and any published work that is in relation to digital forensics.

Secondly, the research will be using case studies to study and understand AI-Powered cyber-attacks, as well as try to identify where the current digital forensics readiness models failed or lacked. The case studies for the research will focus on deepfakes, AI-Powered phishing as well as autonomous malware.

Thirdly, to gain more understanding as well as broader and insightful knowledge, I will be performing interviews and surveys. Gaining insights from professional cybersecurity specialists who specialize in digital forensics can help with understanding the current models and help with understanding in a real life perspective.

Lastly, the research will propose a digital forensic readiness model that includes the identified gaps and aims to address the current issues. The goal is not high detection but rather to show improved readiness and response. The results will be based on how well the model identifies, responds to, and preserves evidence during those scenarios

5 Planning

The following is a bullet point timeline outlining the dates that certain aspects of the research will be complete by. Assignment 1 (Research proposal Peer review) - 06th April 2025

Peer reviews deadline - 17th April 2025

Assignment 2- Research Proposal- (Marked by supervisor) - 29th April 2025

Research report Submission (Marked by supervisor) - 12th October 2025

Research report Poster day

1. Poster submission - 17th October 2025
2. Poster presentation - 24th October 2025

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